

# JAINDU SAMARANAYAKE

+94 76 47-444-88 ◇ devnaka6912@gmail.com ◇ LinkedIn ◇ GitHub ◇ Engineering Portfolio

## EDUCATION

**Bachelor of Science in Electronic and Telecommunication Engineering** March 2024 - Present  
University of Moratuwa, Sri Lanka CGPA: 3.6/4.00

**Upper Secondary Education - Physical Stream** June 2020 - December 2022  
Central College, Kuliyaipitiya [G.C.E Advanced Level Examination - 3A's]

## TECHNICAL SKILLS & INTERESTS

|                                |                                                                  |
|--------------------------------|------------------------------------------------------------------|
| <b>Programming &amp; Web</b>   | Python, Java (Spring Boot), C++                                  |
| <b>Hardware &amp; Embedded</b> | ESP32, Raspberry Pi, STM32L0, Arduino Mega, DE0-Nano (FPGA), I2C |
| <b>Engineering Software</b>    | SolidWorks, Altium, Intel FPGA (Quartus), MATLAB/Simulink        |
| <b>Domains of Interest</b>     | Machine Learning, Automation, Robotics, Deep Learning, LLMs, IoT |

## PROJECTS

**Dengue Patient Monitoring System (Team NEXORA) | SLIoT & SPARK Challenge** November 2025 - Present

- Developing an automated medical monitoring system utilizing a centralized hub and sensor networks.
- Optimizing current diagnostic workflows by integrating Machine Learning for data analysis to reduce human interaction and improve patient outcomes.

**Autonomous Robotics (SLRC 2026 & Robot Design) | Academic & Competition** July 2025 - Present

- Handled the virtual simulation and automation components for the Sri Lanka Robotics Challenge (SLRC) 2026 robot.

**Smart Water Management System | Academic Group Project**

- Developed an automated smart water meter integrated with a web application for real-time data monitoring and resource management.

**Laboratory Database Management System | Academic Project**

- Developed a full-stack laboratory management platform utilizing Java Spring Boot for the backend API and developed the frontend admin dashboard.

**Enterprise Network Design | Academic Assignment** March 2026

- Designed the backbone network architecture for the University of Moratuwa and the internal network for the ENTC department building.

**Hardware Product Development (Sound Regulation IoT) | Curriculum Projects** February 2025 - November 2025

- Designed an IoT device and an analog circuit to divide sound inputs into bands and regulate microphone feed, including custom PCB layout and enclosure design.

## POSITIONS OF RESPONSIBILITY & EXTRACURRICULAR

- **Committee Member, IEEE Robotics and Automation Society (RAS)** November 2025 - Present  
Organized MAZEX 1.0, a prominent Micromouse competition and associated technical workshop series.
- **Organizing Committee Member, ENTC Career Fair '26** January 2026
- **Organizing Committee Member, Sri Lanka Robotics Challenge** April 2025
- **Mentor, Soyuru Sathkara Seminar Series** January 2025